Multimedia Content Management within Multimedia Information Systems for eLearning

Thomas Kleinberger, MCS
Content Management Systems Division
tecmath AG, Sauerwiesen 2, 67661 Kaiserslautern, Germany
kleinberger@cms.tecmath.com

Dr. Andreas Holzinger, PhD, MSc, MPh, BEng, CEng, DipEd
Institute of Medical Informatics, Statistics and Documentation (IMI)
Graz University, Engelgasse 13, 8010 Graz, Austria
andreas.holzinger@kfunigraz.ac.at

Paul Müller, Prof., PhD
Department of Computer Science
University of Kaiserslautern, Paul Ehrlich Strasse, 67653 Kaiserslautern, Germany
pmueller@rhrk.uni-kl.de

Abstract: One of the greatest challenges facing content providers, especially teachers using multimedia information systems (MMIS) for learning purposes, is how to manage complex semantic content. But management of digital content (classifying, storing, retrieving) in general, despite the work over the past 50 years, remains still a major problem. In our paper we describe, what a integrated Content Management System (CMS) must provide to cover this problem for various groups of teachers in eLearning.

Keywords: Multimedia, Content Management, Information-Systems, Metadata, eLearning

1. Introduction

1.1. Media data – Metadata - Content

Ever since Multimedia (images and sound) is available in digital format – and thus as data – it has become common use to distinguish between **media data** – image and sound data – and Metadata describing the actual media. According to a standardization suggestion by the Society of Motion Pictures and Television Engineers [5], image and sound data are called **essence**, while all descriptive data is being called **metadata**.

"Meta" in this context means as much as "about" – indicating that metadata contains information *about* media data. Metadata are for example information about the title of a learning theme, its length and its content. Curtis, Foster & Stentiford [2] describe how metadata can look like, for example in the broadcast business.

More common metadata is described as being "data about data", which provides a means by which the multimedia's semantics can be described in a structured fashion for use by various applications. According to Dorner [3] metadata is standardized information to describe digital information resources. Metadata created only for use by a single application goes against searchability, extensibility, reusability and scalability. Thus, a standard is necessary that everyone conforms to for the exchange and use of media [12]. But even this may not be enough, especially for interoperability issues e.g. spoken language independency.

However, essence and metadata together are summarized under the collective term of **content**. In eLearning-facilities this content can include network accessible resources, including text, images and multimedia documents as well as pointers to resources (URLs) and information about resources like structuring, presentation and ordering information.

Content varies in different application domains, here some examples:

- Broadcast: Analog and digital audio/video material residing on video tapes or an videoservers. The essence are typically videostreams in different coding formats, different compression schemas and different bandwidths. Metadata in this application domain describes a vast amount of object related and organisatorical issues: Technical parameters of the essence, subjects, picture and story descriptions as well as administrative information about television programs or the production process.
- Web: Static and dynamic web pages created by various applications, scripts, servlets connected to databases and including links and multimedia objects like pictures, graphics, audioclips and videoclips.
- eLearning: Computer based learning material such as applications, simulations, presentations, also web pages or audio/video content in conjunction with computer supported cooperative (CSCW) learning features.

1.2. How can content be managed?

According to Passerini & Granger [18] a fully developmental model for designing courses consists of five main phases: Analysis – Design – Development – Evaluation – Delivery. Within the **analysis phase,** teachers focus on learning objectives and instructional context aimed for a particular target group (e.g. for students of medicine and professionals in a medical virtual campus). The preferred strategy for developing the instructional content has to be outlined in the **design phase** [7]. This phase not only includes the definition of a flexible lesson layout, but also elaborate teaching activities within the content research. The **development phase** generates the lesson materials (semantic content) based on the hypermedia design phase specifications and are integrated and synchronized in the software applications chosen for production. This includes the design implementation and the technical production of the content (digitized content) in multiple formats. The **evaluation phase** includes product review during and after production (formative and summative evaluation [8]). Finally the delivery phase includes technical considerations of delivering the content.

In short we can conclude that gathering and managing content involves five important steps:

1) identification of potential resources;
2) assessment of the value of the suggested resources;
3) creation of metadata;
4) monitoring the value of the resource; and
5) overall management and maintenance.

First of all there is to identify potential, suitable and relevant resources for inclusion in multimedia information systems. Therefore the proposed content must be created, identified and evaluated by skilled educators, in our example medical professionals (MD's). This processes of identifying and evaluation education materials could be organized centrally.

Secondly, content assessment involves verifying the standard of the content against content standards and should be undertaken by a content assessor. It will be up to each clinical department to nominate their authorized assessors: it could be any teacher in the system/sector, someone at a system level or nominated individual teachers, professional associations that comply with the content assessors criteria.

Thirdly, the creation of metadata is essential. It is expected that the proposed mandatory metadata will be developed by the initial suggester of the resource. The highly desirable and optional metadata will increase the accessibility of the resource to teachers and students but will require more expertise and will be more costly to develop and maintain. This is likely to only be completed for more valuable and sustainable resources.

Fourthly, monitoring the value of the resource is necessary. In a dynamic web environment, it will be necessary to monitor the value of sites. This will include regular visits to sites that have been included to verify that they are still valid for inclusion on the campus and links outside are following up changes to site addresses.

Last but not least the further management and maintenance of the databases are required. Once information has been added to the databases the database management tasks would include:

- dealing with duplicate records;
- automatic indexing of identified web documents;

- monitoring the consistency and application of metadata to resource;
- identification and appropriate treatment of resources that have moved or no longer available; and
- reporting on status of the database and its use.

1.3. Why is metadata needed?

Traditional search techniques are user directed, manual and often time consuming tasks. The reason for creating metadata, from the perspective of the provider of learning material, is to improve the possibilities of document retrieval (especially the search precision) as well as to support control and management of collections. But there is still another problem remaining: Digital documents with their abundance of different formats and control measures might not always be usable directly by everyone. The format might be unreadable, the content can be encrypted, even prohibited or only permitted after payment. The document might be large, difficult or time consuming to access etc. Especially if multimedia documents are used in education like audio and video, metadata is extremely helpful in managing the multimedia learning content. A solution for all those cases could be metadata which supports the users decision process. Such metadata, necessary to describe educational material homogenous, could be e.g. language, title, catalogue numbers, date, format etc, and in the case of audio and video a time-based documentation of the audio/video image content on a layered basis depending on the documentation focus.

As these multimedia objects typically get larger and larger, a manual metadata documentation becomes very expensive and is not practical anymore. In these cases Content Management Systems (CMS) can support automatic extraction of metadata with audio/video analysis processes that e.g. separate clips into scenes or extract relevant key-frames [12].

2. Multimedia Content Management

2.1. The need for Content Management

Modern computer based teaching and learning systems not only present knowledge and information in words and pictures. They include simulations and animations and offer individually designed and interactive learning facilities. New multimedia technology in conjunction with the use of the world wide web are an ideal basis for reforming and modernizing education: We call this eLearning.

As computer based education becomes increasingly available, the amount of online learning material will increase consequently more and more. New learning material will be build up more and more of multimedia and hypermedia content for easier illustration [6], [7], [14], [21]. With the focus on multimedia and hypermedia, the growing amount of learning content therefore has to be managed to reach the following basic goals:

- managing a large amount of interactive multimedia and hypermedia learning material;
- keeping the overview about the whole content used in learning programs;
- storing and archiving learning content with a centralized access;
- preserving physical quality of the learning content;
- making the learning content reusable for further processing in new learning content creation;
- providing functions for searching, content browsing at the site of the learner over the world wide web;
- allowing retrieval and delivery directly to the learner over the world wide web.

The general solution to reach these goals is to separate the objects to be managed into smaller pieces according to the principle “divide & conquer”. Separation can of course be done by breaking larger content pieces into smaller ones. But Content Management goes one step further and separates additionally the structure and presentation information from the essence information.

- Structure: Definition of elementary information objects, their structural organization and their ordering
- Presentation: Formal description to represent (display, layout, execution,…) on a possible output medium
- Essence: The content of the elementary information objects, which are organized according to the structural definition. Storage management of this essence is of important significance, because the metainformation has still to be assignable to every elementary essence object as well as to every higher structure.

The separation of structure, presentation and essence the main principle of Content Management.

2.2. Media Data – Essence – Metadata

How does content look like in the application domain we want to look at: eLearning? Here we deal with texts, pictures and graphics, with hypertexts, multimedia and hypermedia but also with active objects like applets, servlets and scripts because learning mostly is still an interactive process. All these media objects, media types and their combination can be summarized in the term essence, while all descriptive and administrative data about essence and it's structure and presentation information accordingly are the metadata.

Typically the metadata is carried together with the essence to provide easier searching and access to the essence objects. Especially in the case of active objects or multimedia objects, these objects are coded in a way which is not indexable or searchable in an easy way. Often there are logical connections between the essence and the metadata. For example, an abstract may relate to a particular time span in a video material, and a usage restriction will usually also apply to a particular time span in a video material.

2.3. Typical characteristics of Content Management

The effective management of content follows the principle of separating content in structure, presentation and essence. Metadata will be used to keep the relationship between these three parts.

Structure can for example be the:

- Definition of essence structure
- Description of essence parts
- Relationship to other content objects
- The hierarchy in an higher essence structure
- Ordering of essence objects
- Different representations of essence objects (e.g. different formats, different codings)

Presentation can be the:

- Definition of essence representation depending on the users environment and way of access
- HTTP presentation in a web browser
- Interpretation of style sheets (XSL) for HTML web pages or XML data
- Streaming for audio/video objects with varying bandwidths
- Transfer methods like download for multimedia data to other destinations for further processing

Essences are:

- The data part of the content that is presented to the viewer or transferred to a destination system,
- which is structured according to the structure information and presented with one of the presentation methods
- and typically stored in databases, repositories or file systems.
- E.g. digital audio/video, elements from websites, eLearning material

2.4. Content Management Systems

Content Management Systems have to implement the general principle of separating content into structure, presentation and essence. According to Kleinberger & Müller [11] the basic features of Content Management in eLearning can be summarized as follows, to support the separation principle:

- *Creation*: The complete process of authoring and production of content. This includes typing, drawing, painting, recording, digitization, encoding and storage of essence and metadata.
- *Registration*: Importing content into a Content Management System. This can be an automatic or manual import. By registering, content gets known to the Content Management system, unique identifiers will be associated and automatic processes for classification and indexing of the content get triggered.

- *Classification*: All kinds of intellectual or automatic analysis and classification of imported content. The intellectual analysis and classification can be supported by appropriate representation forms and documentation tools. The automatic analysis and classification can be supported by methods of cut recognition, key frame extraction, audio classification, speech recognition, face recognition and other kinds of analysis methods suitable for metadata creation. Classification can also be performed manually by adequate tools for documentation and arrangement.
- *Search*: Searching for content either by metadata, which is linked to the essence, or directly by the essence. Typically full text search or similarity search are the commonly used methods. Content based search can be enhanced by filtering, personification, interpretation or methods of case based reasoning and complemented by attribute based searching.
- *Retrieval*: Retrieval of found content by transfer, play, browsing, streaming, display, distribution, download, export or connection.
- *Modification*: Changing of essence or metadata with modification and documentation tools. These tools are typically the same tools as the tools used for classification.
- *Storage Management*: Content preserving and archiving in file systems, HSM systems or archives, repositories and databases, automatic or manual format conversion for retrieval through different connections, i.e. different bandwidths in networks.
- *Workflow management*: Methods for combining and sequencing the basic functional and management features in a learning workflow.
- *Additional blocks* for special usage areas: Additional functions in some usage areas like rights management, e-commerce and billing & accounting which affect all building blocks above. This can also be the connection of external devices for audio and video or the connection of legacy systems.

Each of these basic features will be represented in a Content Management System architecture more or less in functionalities accessible by users (user applications, web interfaces), middleware layers (distribution abstraction), system services (full text engines, indexing services) or supporting services (like repositories and databases).

Thus the term **Content Management System (CMS)** can be defined as a *software system which provides functions and tools for the tasks and functions on essence and metadata* as described above. In eLearning the essence consists of the learning material in different variations, i.e. hypertext, multimedia, hypermedia, text, graphics, pictures but also active components. The metadata is the associated descriptive data related to the essence.

How does a real Content Management System look like? Are there any reference architectures for Content Management Systems or metadata representations? Several suggestions exist, especially for the application domains we have mentioned in the beginning: Broadcast, web and eLearning. We mention a few proposals for metadata representation and to present two reference architectures for the eLearning application domain in a little more detail in the following sections.

3. Metadata representations

Metadata is information about an object and as the number of objects increases, the lack of information or metadata about objects produces a critical and fundamental constraint on our ability to discover, manage and use objects. Only standards might address this problem by defining a structure for interoperable descriptions of learning objects. According to Wantz & Miller [23] there are four remarkable efforts concerning eLearning:

- 1) Dublin Core [4]
- 2) GEM (Gateway to Educational Materials) [22]
- 3) Warwick-Framework [16] and the
- 4) IEEE Learning Objects Metadata (LOM) [17]

A more detailed comparison of these efforts can be found in [9].

4. Reference Architectures for eLearning

4.1. The Learning Technology System Architecture (LTSA)

A well known approach for a teaching and learning architecture has been proposed by the Learning Technology System Committee (LTSC) of the IEEE [17]. The primary objective of LTSC is to specify standards and references for every subject concerning CAE (Computer Aided Education). In 20 working and study teams, grouped in the 5 higher categories General, Learner-Related, Content-Related, Data and Metadata and Management Systems, almost any theme is addressed, that is related to CAE.

The main proposal, that is of interest for our subject, is the "Architecture and Reference Model", developed in the working and study group P1484.1. The proposed description of a "Learning Technology System Architecture" (LTSA) is actually available in version 6.00 (draft) and specifies a common reference model and architecture for component based CAI (Computer Aided Instruction) systems, especially including the needs of intelligent learning environments and intelligent tutoring systems. In diverse companion specifications several other models such as learner models, task models and student models are developed, that can be components of larger CAE systems, too. The general architecture of LTSA consists of five layers (Figure 1):

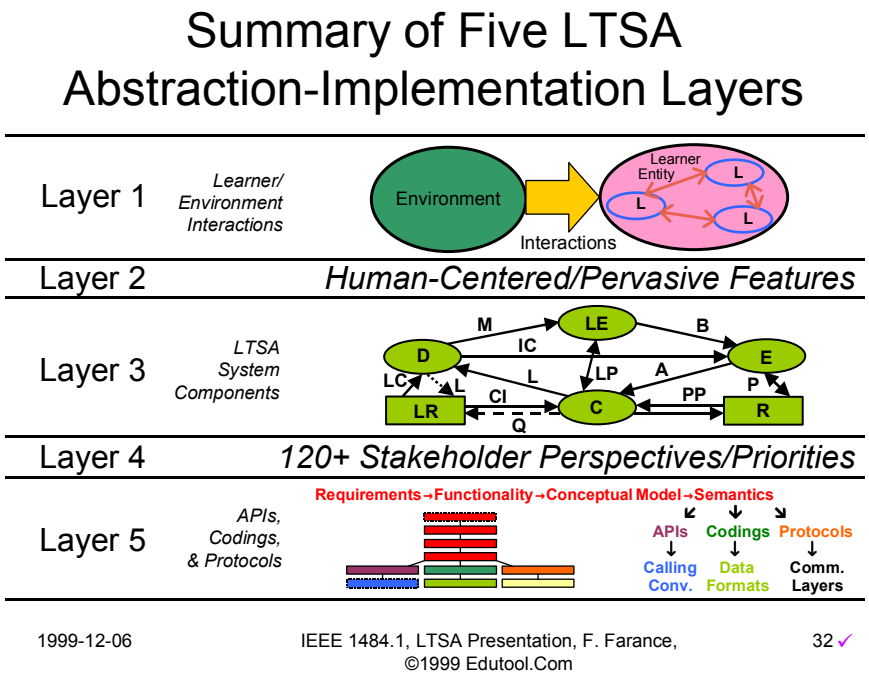


Figure 1: The five LTSA abstraction-implementation layers

- Learner and environment interactions: Interaction of the learner with the CAE system through different communication techniques.
- Human-centered features: Human aspects of the learning process mapped to learning technology.
- System components: Mapping of the human aspects to system components.
- Stakeholder perspectives: A variety of different viewpoints dependent on different subsets of systems components.
- Operational components: Technologies for implementation of components, protocols and interfaces as identified in the stakeholder perspectives.

As we can see, the environmental requirements on teaching and learning architectures map to the first two layers of LTSA and build the functional requirements for the actual system components in layer three (Figure 2). The fourth

layer deals with the various views on the system architecture depending on applications subdomains, detailed requirements and design issues while the bottom layer five supposes mappings for implementation.

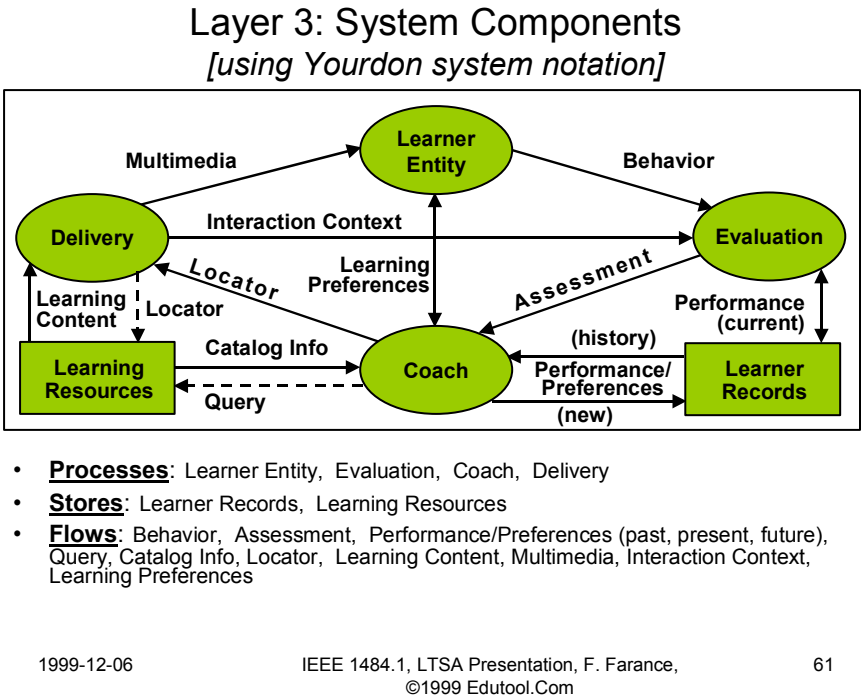


Figure 2: The LTSA layer 3 system components

LTSA offers a classical perspective of technology for CBT in the sense that the conceptual model is designed as a dataflow model, describing stores, processes and dataflows for the eLearning domain. The LTSA working group states that the architecture should be descriptive and not prescriptive. This means, that the proposed architecture does not claim to be a building plan for a real system architecture nor lists any features that have to be implemented. Instead it should provide a possibility to compare a real system architecture to a reference. In any case the design of a real system architecture which supports the eLearning workflow will vary greatly for every application subdomain in eLearning (cf. the amount of stakeholder perspectives in LTSA). For this reason we look at LTSA as a workflow reference model, helping us to identify the various possible processes, entities and flows in eLearning systems.

4.2. Telelearning Reference Architecture

Another reference architecture for eLearning is proposed by F. Milius [1], [13]. This architecture is divided into four component layers (Figure 3):

- Campus Components,
- Knowledge Components,
- Production Components and
- Database Components.

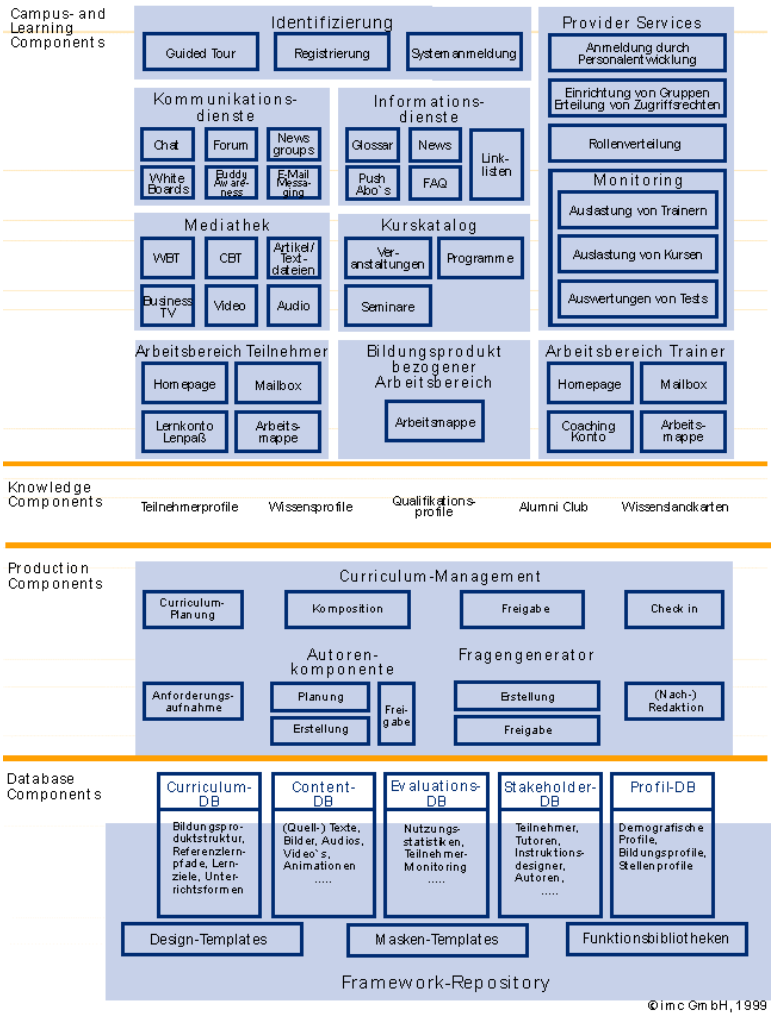


Figure 3: The Telelearning Reference Architecture

Campus Components: Campus Components are services components, that offer their services without direct reference to a specific educational product, for example identification services or information services (yellow pages, hint list, etc.). On the layer of campus components communication and collaboration services like chat or discussion lists can be found, too.

Knowledge Components: The private workspace of each user, where each user can manage his personal data. Users can be learners or teachers/tutors. A public workspace on this layer provides a communication and collaboration platform between learners and teachers and, in turn, between learners and learners. The public workspace allows for a realization of open and closed user groups.

Production Components: Production components support the processes between instructional designers of a educational product and the developers of the media components. A difference is made between components for Curriculum Management and components for Content Management.

Database Components: The database components integrate the contents of the eLearning applications. A logical difference between informational data (stakeholder database, profile database) and educational products (curriculum database, Content database, evaluation database) is made. Every data is described semantically and syntactically in a repository, so that applications can be modified and extended in a flexible way according to the specific demands of concrete learning systems.

In comparison to the LTSA reference architecture, the Telelearning Reference Architecture proposes a more detailed view on a possible real system design consisting of system components, stores, features and functions. Unlike LTSA it doesn't specify processes and flows, neither does it propose operational components like APIs, codings, protocols or interface mappings.

5. Repositories for Multimedia

As more and more educational content becomes digital and will be delivered and presented electronically some questions arise such as how to administer the continuously growing amount of electronic content or how to manage the large variety of electronic presentation techniques and storage formats. Each author and provider of electronic learning content will be forced to build up a smaller or larger archive, in which the whole electronic learning content will be stored. The main purpose for this need is the wish to preserve the technical quality of the content, to keep an overview of all content available and of course to reuse already created content in new productions. But how can such a content management be done? In order to find an answer, we have to divide the problem how the management of electronic content can be realized into a few smaller problems such as:

- How to document content it in the right form, so that something can be searched for by using this documentation
- How to store content in the right way, so that it can be found again
- How to store content in the right format, so that some parts of it or the whole content can be retrieved and reused
- Where to store content, so it can be accessed instantaneously for retrieval or reuse

To solve this storage management problem, a general management of content storage has to be introduced. One proposed solution can be realized as a part of a Content Management System [11], which is responsible for the management of digital assets, the storage of digital media, the access to the stored media and the support of retrieval, reuse and production processes.

In consequence a Content Management System has to provide a storage management as a core component which stores all the objects the content in eLearning is built up from: essence (multimedia and hypermedia) and Metadata (structure and presentation information). There have to be different storage areas according to different requirements for the objects stored in these areas. Metadata and essence objects have different characteristics such as access method, object size, access time etc. and therefore need different treatment.

We can summarize the requirements for a storage management for hypermedia and multimedia eLearning content in the following terms:

- Storage of multimedia eLearning content
  - Store content, consisting of different essence and metadata objects
  - Store multimedia documents consisting of multiple formats and multiple physical objects
  - Provide different storage areas with different access methods for different kinds of essence and metadata
  - Examples of content: audio, video, animations, text, pictures, hypertext, hypermedia
- Flexible access to the stored documents for retrieval and reuse
  - Allow proper documentation of essences in a generalized manner
  - Provide indexing mechanisms to extract metadata and semantic information automatically
  - Provide advanced search capabilities for essences using extracted metadata information
  - Offer filtering and scaling methods for retrieval according to available bandwidth or location
- Integration in teaching and learning environments
  - Decentralized implementation with transfer and replication mechanisms
  - Remote access facilities with broker mechanisms
  - Integration of rights management

How do the basic architectures look like? A storage management for eLearning content can base on three kinds of storage management architectures, which are currently state of the art:

- Database Systems
- Data repositories
- Intelligent data management

The advantages and disadvantages of these basic architectures are discussed in [19]. In the following section, we describe one possible solution of a multimedia storage management for the application domain eLearning, which bases on one of these basic architectures.

5.1. Application domain eLearning: The L3 project

The L3 project (L3 stands for Life Long Learning [15]) is a project founded by the German government with the goal to create a technical and organizational infrastructure for a lifelong further education. The solution approach emphasizes two different points: On the one hand the creation of an educational infrastructure which uses new media efficiently and can be used by all interested people, independently of there education or social position. On the other hand the development of organizational structured and economic business models with which the developed infrastructure can be operated at a medium term.

The central component of the educational infrastructure is a multimedia repository [12] which manages online educational content, especially multimedia content like audio and video. The educational multimedia content consists of learning objects which can be structured hierarchically in courses and lessons. The actual relationships between the learning objects are defined in a course structure definition which is represented in XML. Metadata for the learning objects is represented in a XML notation of the IMS metadata specification version [10] which is based on the IEEE LOM definition with some changes and extensions. The LOM definition was extended with the structures necessary to store all metadata information not already included, i.e. the course structure. Media objects are referenced out of the metadata objects to physical media objects stored in the multimedia repository.

5.2. Multimedia Repository Architecture in L3

The proposed multimedia repository architecture in L3 takes care of all requirements for a multimedia storage management given above. It provides different interfaces to both end user applications and additionally to components of a Content Management System which is embedded in the learning environment. The basic features of the multimedia repository provided by its internal functions and interfaces are:

- The management of a large amount of interactive multimedia and hypermedia learning material
- The provision of a centralized access to all stored content
- The preservation of physical quality of the learning content
- The provision of functions for content based searching, content browsing, retrieval and reuse

To fulfill all requirements for a multimedia storage management, the internal structure of the multimedia repository has been separated into dedicated components, each of them realizing certain functions and providing special interfaces. A general classification into two component groups can be made due to the different load profiles the multimedia repository is faced with for Metadata and Essence.

On the one hand the multimedia repository has to do the work of a classical database system. It has to handle a great amount of user load and to answer a lot of user inquiries with adequate computing power. This can be compared with a typical database load profile. On the other hand a very performant storage system is required, providing capability of handling large multimedia objects, managing a large storage area (typically in a file system) and providing high data throughput over network and audio/video interfaces. This can be compared with a typical file server profile, here with auxiliary interfaces such as streaming servers and audio/video encoders and decoders.

In classical information technology both load profiles, database and file servers, are handled by different system configurations and software solutions. For example database servers are typically well equipped with a huge main memory and computing power (e.g. multiprocessor) whereas file servers are well equipped with a performant hard disk system (e.g. Hardware-RAID-Controllers, FC-interfaces, HSM-Systems, Caches) and fast networking interfaces.

To realize a repository for multimedia objects, both load profiles have to be addressed. Hence a multimedia repository is internally organized into at least two different components according to the classification: a repository addressing the database load profiles and a media server addressing the file server load profiles. Figure 4 summarizes the basic features of the two components:

	Repository	Media Server
Load Profile	Many accesses, Computing intensive	Less accesses, High I/O Throughput
Software	Database, Fulltext Search Engine, Web Server	Video-, Audio- , Web-, File Server
Requirements	Computing Power, Memory	Performant Storage, High I/O Throughput

Figure 4: Requirements

Figure 5 shows the general separation of the two components repository and media server in the multimedia repository.

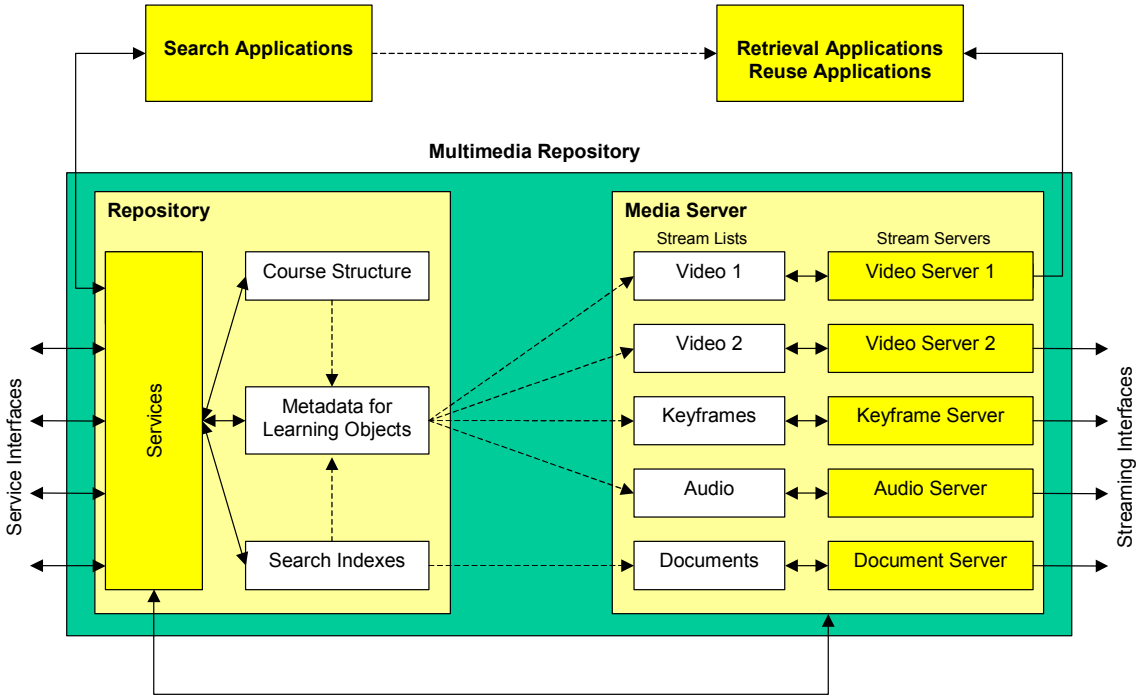


Figure 5: The L3 Multimedia Repository and Media Server

The multimedia repository stores all the metadata information about the educational multimedia content. This is the descriptive metadata for the educational content like author, theme and any other metadata information like special attributes for certain domains. Because an educational content object is mostly embedded in an environment, compound documents consisting of content objects can be structured into a course. The course structuring information is stored in the multimedia repository as Metadata information, too. All Metadata information can be queried by XQL or searched for with a fulltext search engine. The multimedia repository provides adequate services and interfaces for these functions.

On the media servers side an essence manager is responsible for the management of the media objects the learning object refers to. The essence manager includes a metadata management which stores the technical metadata about the physical media objects. Since a media object can also consist of more than one physical media object (e.g. files), the essence manager furthermore manages containers, which aggregate physical media objects into material lists which are then referenced by media objects. Physical media object can be stored in different formats and different bandwidths, e.g. RealVideo, MPEG-1 or MPEG-2. Formats are distinguished in stream lists which point to the location of the physical media objects as files in a file system. Retrieval access to a physical media object can now be gained through a corresponding stream server. Figure 6 shows a more detailed view on the repository and media server, in which the media servers components stream server and essence manager are detailed.

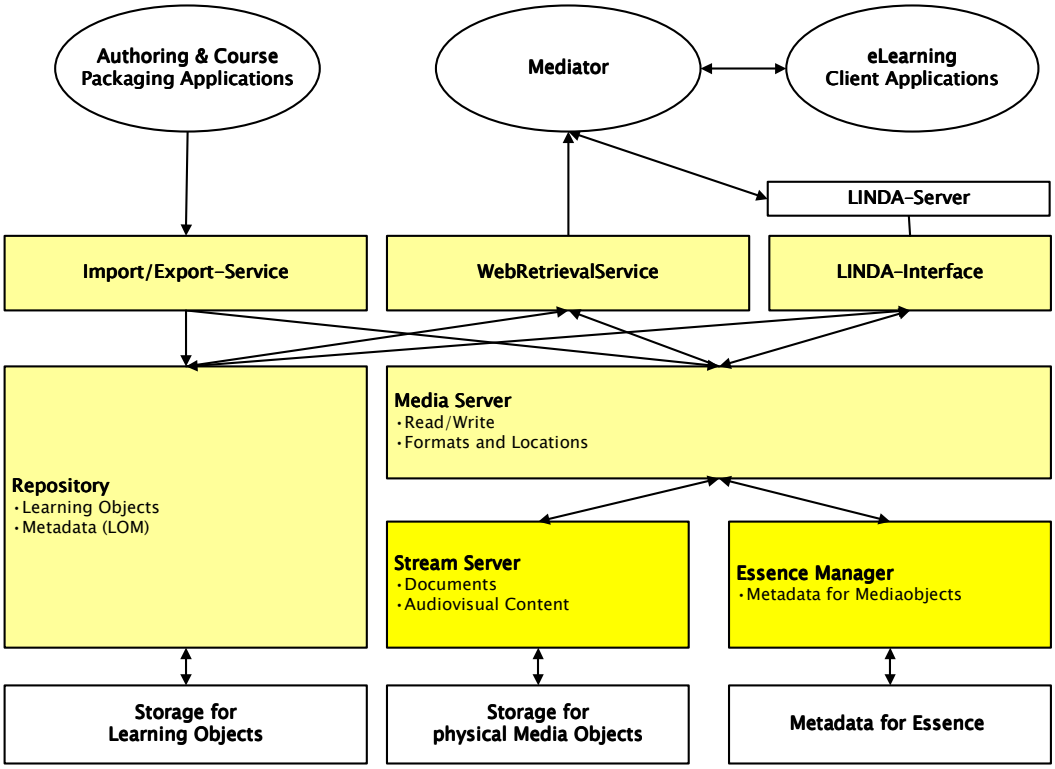


Figure 6: The L3 Repository and Media Server System Components

There are different stores for the learning objects, the media stores and the metadata for essences. Objects are stored in a database management system, file system, HSM-System or in a storage management system. Various services like import/export services (for importing and exporting complete packed learning objects (courses) including Metadata and Essences into and out of the multimedia repository), a web retrieval service (for retrieving media objects) and a Linda interface [20] (for accessing all administrative and Metadata related functions) are provided to support authoring and course packaging tools, mediators (which support course navigation) and eLearning client applications.

In a typical installation both components, repository and media server are installed on separate computers, each built up of the appropriate hardware components in computing power, main memory, hard disk storage and network interfaces depending on the load profile.

6. Summary

We have given an explanation and definition of the terms content, metadata and essence and discussed why metadata is needed and how multimedia content in the eLearning application domain can be managed with Content Management. Content Management follows the principle of separating content into information structure, presentation and essence. Content Management Systems have a central position within a MMIS in the eLearning application area to manage multimedia content effectively. The core components of Content Management Systems,

a multimedia storage management, data repositories and intelligent data management, can improve the storage and access of multimedia and hypermedia content to gain a common solution, how do document content, how to store it in the right format and right way and where to store it so that a continuously growing amount of varying multimedia content can be mastered.

A few proposals for metadata representations and reference architectures in the eLearning domain were presented, each of them not providing a complete description and solution of the Content Management Problem in eLearning. Finally an example of a project in the eLearning application area, the L3 project, was presented with a possible solution for Content Management on the basis of a multimedia repository.

7. References

[1] Bertelsmann (1999): University development through new media: Conference of the Bertelsmann Foundation Gütersloh, Germany, October 1999

[2] Curtis K.; Foster P. W., Stentiford F. (1999): Metadata – The Key to Content Management Services, BT Laboratories.

[3] Dorner, Dan (2000): Cataloging in the 21st century — part 2: digitization and information standards. Library Collections, Acquisitions & Technical Services, 24, 73 – 87.

[4] Dublin Core Metadata, http://purl.org/DC/index.htm.rdf

[5] EBU/SMTPE (1998): Task Force for Harmonized Standards for the Exchange of Programme Material as Bitstreams, Final Report August 1998

[6] Holzinger, Andreas (2000a): Basiswissen Multimedia, Band 1: Technik, Technologische Grundlagen multimedialer Informationssysteme. Würzburg: Vogel.

[7] Holzinger, Andreas (2000b): Basiswissen Multimedia, Band 2: Lernen, Kognitive Grundlagen multimedialer Informationssysteme. Würzburg: Vogel.

[8] Holzinger, Andreas (2000c): Basiswissen Multimedia, Band 3: Design, Entwicklungstechnische Grundlagen multimedialer Informationssysteme. Würzburg: Vogel.

[9] Holzinger, Andreas; Kleinberger, Thomas; Müller, Paul (2001): Multimedia Learning Systems based on IEEE Learning Object Metadata (LOM), ED-MEDIA 2001 World Conference on Educational Multimedia, Hypermedia and Telecommunications, Tampere, Finnland, June 25-30, 2001.

[10] IMS: Instructional Management Systems Project, http://www.imsproject.org

[11] Kleinberger, Thomas; Müller, Paul (2000): Content Management in Web Based Education. Webnet 2000 Conference on the WWW and Internet, San Antonio, Texas, USA, October 30 - November 4, 2000

[12] Kleinberger, Thomas; Schrepfer Lutz; Holzinger Andreas; Müller Paul (2001): A Multimedia Repository for Online Educational Content. ED-MEDIA 2001 World Conference on Educational Multimedia, Hypermedia and Telecommunications, Tampere, Finnland, June 25-30, 2001.

[13] Kraemer, Wolfgang (1999): Corporate Universities - Knowledge for leadership in 5. IT-Trainings-Kongress, 11.+12. November 1999 in Bonn (incl. F. Milius E-Learning Framework), http://www.synergie-gmbh.de/5it-kongress/themen/t-vortraege/t-v-kraemer_4.ht

[14] Kuhlen, Rainer (1991): Hypertext, Ein nicht-lineares Medium zwischen Buch und Wissensbank, Springer-Verlag

[15] L3: Lifelong Learning – Education as a basic need, Leitprojekt des Bundesministeriums für Bildung und Forschung zum Themenfeld „Nutzung des weltweit verfügbaren Wissens für Aus- und Weiterbildungsprozesse“, http://www.l-3.de (last visited: 30th December 2000)

[16] Lagoze C.; Lynch C.A.; Daniel R. Jr. (1996): The Warwick framework: a container architecture for aggregating sets of metadata. Cornell University Computer Science Technical Report, TR96-1593. June 1996. On-Line at: http://cs-tr.cs.cornell.edu/Dienst/UI/2.0/Describe/ncstrl.cornell/TR96-1593.

[17] LTSC: IEEE 1484 Learning Technology Standards Committee LTSC, http://ltsc.ieee.org

[18] Passerini, Katia; Granger, Mary J. (2000): A development model for distance learning using the internet, Computers & Education 24 (2000) 1-15, pp.

[19] Pazandak, Paul; Srivastava, Jaideep (1997): Evaluation Object ODBMSs for Multimedia, IEEE Multimedia, Vol. 4 No. 3 July-September 1997, Pages 34-49

[20] Schoenfeldinger, W.J. (1998): The Perl-Linda Server, Technical Report Nr. 12, Department for applied Computer Science, WU-Wien, Austria

[21] Steinmetz, Ralf (2000): Multimedia Technology, published by Springer, 3. Edition

[22] Sutton, Stuart A. (1998): Gateway to Educational Materials (GEM): metadata for networked information discovery and retrieval. Computer Networks and ISDN Systems, 30, 691-693.

[23] Wantz, Jay L.; Miller, Michael (1998): Educational Metadata in use. Computer Networks and ISDN Systems, 30, 701-703.

About the authors:



Thomas Kleinberger works at the tecmath AG in Kaiserslautern as a consultant for digital multimedia solutions in broadcasting and eLearning environments. He received his MCS from the University of Kaiserslautern in 1991. His current research interests include digital libraries, multimedia databases and repositories, multimedia content management for broadcasting, eLearning environments and live long learning. He is a member of the IEEE Computer Society, the AACE, the SMPTE and the German Society for Informatics (GI).



Andreas Holzinger works, teaches and does research at Graz University in Information-Systems – Multimedia – Human-Computer-Interaction – Usability – Internet/ Intranet – Intelligent Tutoring Systems. He holds a CEng in electronics, a BEng in communication engineering, a MSc in Physics, a MPh in Media and a PhD in Cognitive Science. He is a Member of the ACM, the IEEE Computer Society, the AACE, the German Society for Informatics (GI), the German Society for Psychology (DGP), the German Society for Media in Science (GMW), and board Member of the Austrian Computer Society (OCG). He is national multimedia expert in the European Union in the eLearning initiative.



Paul Müller is a professor of computer science at the University of Kaiserslautern where he directs the working group Integrated Communication Systems (ICSY) and the regional computing center. He received his diploma in mathematics from the University of Bochum and his Dr. degree from the University of Ulm. His current research interests are in integrated communication systems like B-ISDN/ATM, distributed multimedia systems and new protocol architectures for QoS networks.